ASL Reader

(American Sign Language Reader)

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Background description:

Over 70 million people use sign language globally. At 500,000, American Sign Language, ASL, is the most used sign language in the world. Within ASL, about 12% to 35% is fingerspelling. Finger-spelling is a form of sign language where one would use one hand to sign one letter at a time to spell out words.

Pain:

Even with an abundance of people using sign language, there is still a huge barrier between those who are hard of hearing and those who are not.

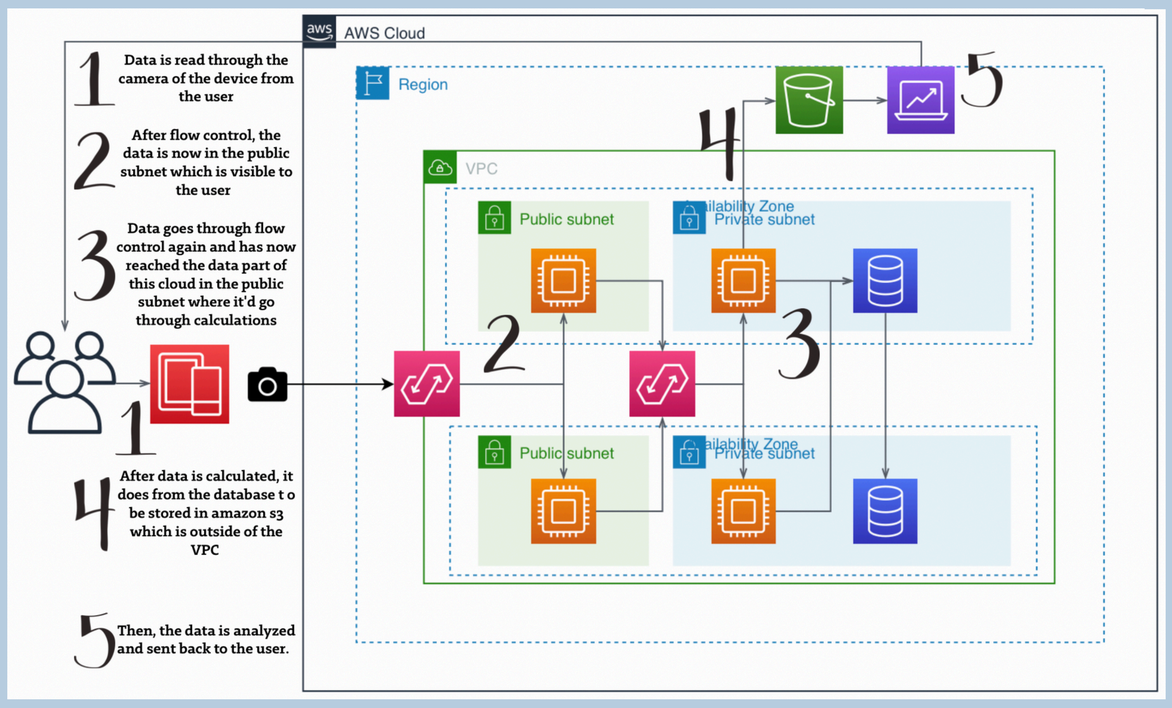
Need:

A tool to help break down the barrier to make using sign language more convenient, also providing those using sign language with more equal opportunities.

User and stakeholders analysis (Role):

- End user : everyone
- Stakeholder 1 : non sign language user
- Stakeholder 2 : mute
- Stakeholder 3 : hard of hearing
- Stakeholder 4 : sign language users

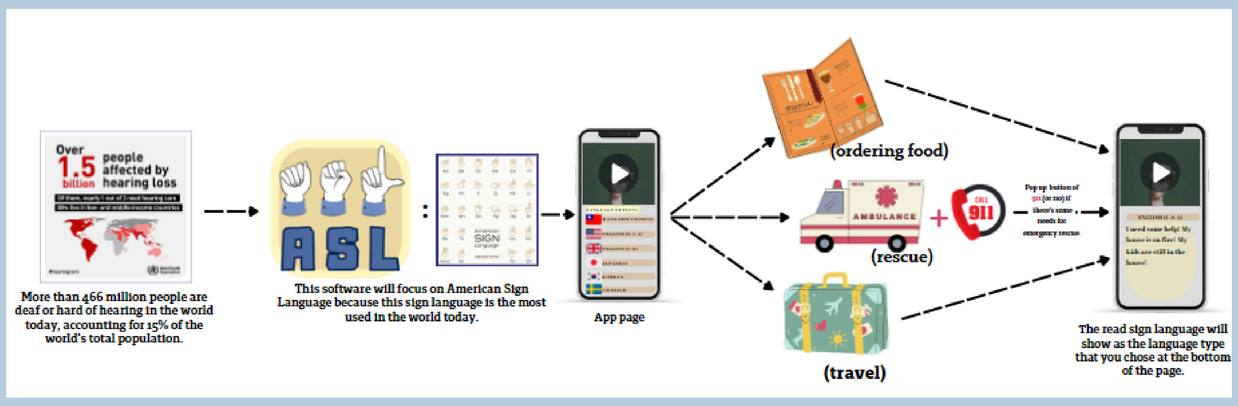
Visual Paradigm:



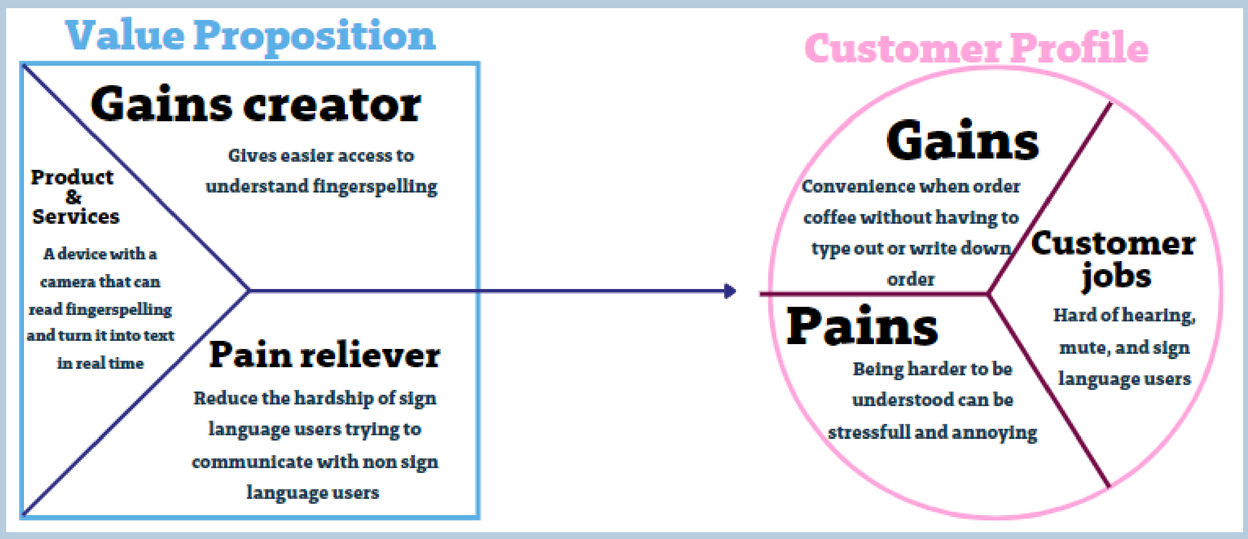
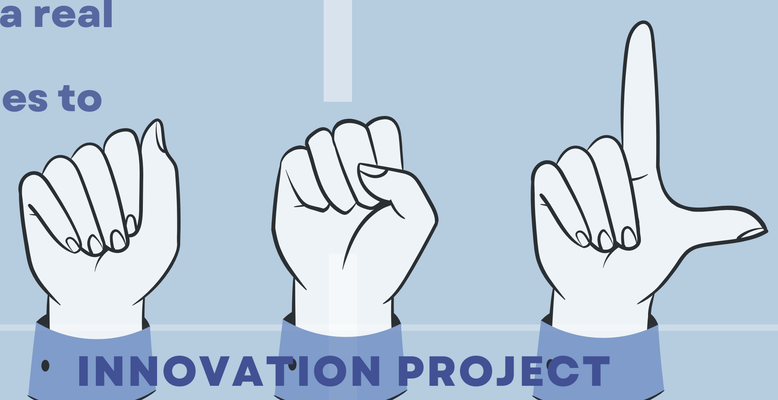
Detailed description of the top solution：

A program is written to recognize the movement of the hand and to analyze the gesture of the hand. Then, the program would examine which fingerspelling hand sign the gesture is closest to. Finally, it would output the letter that corresponds to the said sign for those that does not understand sign language to read.

Schematic diagram：



Evaluation: (WDM/SWOT ANALYSIS)

- **Threats :**
  Typing on a smartphone or writing things down on a piece of paper might be faster an easier to access
- **Opportunities :**
  When a barista is taking orders in a coffee shop
- **Weaknesses :**
  There might be false readings which could lead to possible misunderstandings
- **Strengths :**
  - Creates equal opportunity for those that uses sign language
  - Takes down the barrier between sign language users and non sign language users
  - Turns sign language into a part of life like a real language
  - can translate final text into other languages to save time

Value proposition canvas :